

RIDGETOWN, ON, Canada

WMO: 713070

Lat: 42.454N

Lon: 81.886W

Elev: 675

StdP: 14.34

Time Zone: -5.00 (NAE)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.6	3.2	-8.9	3.5	-1.8	-2.6	4.9	3.7	25.6	27.5	23.5	26.7	6.5	230	0.531

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.1	86.8	74.1	84.2	72.5	81.6	71.0	76.6	83.7	74.8	81.3	73.2	79.2	10.4	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.6	132.5	80.7	72.8	124.7	78.8	71.2	118.0	77.1	40.5	84.2	38.8	81.3	37.3	79.1	85.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
22.3	19.7	17.6	-9.1	91.4	7.4	2.2	-14.4	93.0	-18.7	94.3	-22.9	95.6	-28.3	97.2		
			-9.5	79.6	7.2	2.6	-14.7	81.4	-18.9	82.9	-22.9	84.4	-28.2	86.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	48.0	23.8	24.5	33.8	45.3	57.2	66.2	70.2	68.4	62.7	51.5	40.2	30.4
	DBStd	18.69	11.87	11.20	10.69	8.49	8.60	6.53	5.66	5.44	7.25	8.75	8.90	9.55
	HDD50	3269	814	714	512	190	25	0	0	0	2	88	315	609
	HDD65	6722	1279	1133	966	590	274	62	16	28	127	428	745	1074
	CDD50	2533	0	1	11	50	248	487	627	570	384	135	20	0
	CDD65	514	0	0	0	1	33	99	178	133	60	10	0	0
	CDH74	3921	0	0	0	13	266	745	1483	952	414	48	0	0
	CDH80	884	0	0	0	0	54	192	373	170	87	8	0	0
	WSAvg	8.1	9.9	9.4	9.2	9.8	8.3	7.2	5.8	5.4	6.0	7.6	8.9	9.4
	PrecAvg	36.1	3.0	2.4	2.4	3.0	3.3	3.1	3.3	2.9	3.2	3.0	3.2	3.0
Precipitation	PrecMax	47.2	4.6	4.9	4.8	4.9	6.0	6.2	6.5	5.2	8.4	6.1	5.8	5.8
	PrecMin	28.3	1.2	0.7	0.9	1.0	0.9	1.2	1.2	0.5	0.7	0.8	1.2	1.2
	PrecStd	5.6	0.9	1.1	1.1	1.0	1.3	1.3	1.1	1.1	1.8	1.3	1.3	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.5	57.1	67.1	75.4	85.5	89.0	90.4	88.2	87.1	79.6	65.3	55.9
		MCWB	51.9	52.4	59.4	60.5	71.6	74.8	75.8	75.9	73.8	69.9	57.7	53.0
	2%	DB	49.0	48.6	59.0	69.4	80.7	85.4	87.0	84.1	82.4	73.0	60.5	50.6
		MCWB	47.6	45.4	53.8	58.0	67.6	72.3	74.5	74.1	70.8	64.8	55.3	48.4
	5%	DB	43.3	42.6	54.0	64.2	76.7	81.6	84.3	81.6	78.7	69.3	57.0	46.3
		MCWB	41.4	39.7	49.8	55.1	65.7	70.4	73.0	72.0	68.8	62.9	52.5	44.1
	10%	DB	39.2	37.9	49.2	59.4	71.6	78.2	81.4	79.3	75.4	65.9	53.9	42.4
		MCWB	37.7	35.7	45.0	51.5	62.5	68.0	71.1	70.5	67.4	60.3	50.1	40.5
	0.4%	WB	52.2	52.9	60.7	63.5	73.6	77.5	79.7	78.4	75.7	71.4	60.2	53.1
		MCDB	53.3	56.3	66.7	72.0	81.5	86.0	85.9	83.0	84.1	79.0	62.7	55.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	47.9	46.0	54.8	59.8	70.0	74.7	76.9	76.1	73.1	67.5	56.5	48.8
		MCDB	48.8	48.6	57.7	66.7	78.1	81.9	84.1	81.7	79.4	70.9	59.7	50.3
	5%	WB	41.9	39.7	50.1	56.4	67.1	72.5	75.0	74.1	71.0	63.8	53.3	44.3
		MCDB	43.1	41.7	53.2	63.1	74.2	79.1	81.5	79.3	76.1	67.9	56.2	45.9
	10%	WB	37.5	35.8	45.4	52.7	63.9	70.1	73.2	72.4	69.0	60.7	50.5	40.7
		MCDB	39.0	37.8	48.6	58.5	69.8	76.3	79.4	77.1	73.6	65.2	53.6	42.3
Mean Daily Temperature Range	5% DB	MDBR	13.3	14.1	14.7	18.0	18.9	18.5	19.1	18.8	19.2	17.0	13.9	11.2
		MCDBR	14.6	17.9	20.8	24.9	22.5	21.3	20.4	19.6	20.9	19.3	17.0	13.7
		MCWBR	14.0	15.9	16.0	16.5	12.9	11.7	10.7	11.2	11.3	11.8	13.4	12.9
	5% WB	MCDBR	14.4	16.7	19.3	22.0	20.4	19.1	17.8	17.3	18.1	17.8	15.9	13.4
		MCWBR	14.2	15.3	16.4	16.2	12.9	11.7	10.3	10.6	11.1	12.3	13.4	13.0
Clear-Sky Solar Irradiance	taub		0.302	0.324	0.344	0.389	0.415	0.436	0.428	0.409	0.377	0.362	0.323	0.300
	taud		2.393	2.307	2.340	2.271	2.248	2.225	2.271	2.309	2.386	2.403	2.471	2.468
	Ebn at Noon		266	277	285	279	273	267	267	269	271	260	256	258
	Edh at Noon		25	32	35	41	43	44	42	39	34	29	23	22
All-Sky Solar Radiation	RadAvg		466	657	1105	1466	1752	1987	1966	1721	1369	872	532	369
	RadStd		43	93	114	135	152	126	117	80	116	73	72	30

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (64 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page